

12. Evaluate $\int_{-1}^0 \frac{dx}{\sqrt[3]{x}}$.

13. What is Cauchy principle value. Find the principal value of $\int_1^1 \frac{dx}{x}$.

14. State Leibniz rule for differentiation of Ramann integrals.

15. State that $\Gamma(p+1) = p\Gamma(p)$ for $p > 0$.

(10 × 3 = 30 marks)

Section B

Answer at least **five** questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Show that the Dirichlet's function :

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases} \text{ is not continuous at any point of } \mathbb{R}.$$

17. State and prove Bolzano intermediate value theorem.

18. Show that the following functions are not uniformly continuous on the given sets :

(a) $f(x) = x^2$ on $A = [0, \infty]$.

(b) $g(x) = \sin \frac{1}{x}$ on $B = (0, \infty)$.

19. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then show that $f \in \mathbb{R}[a, b]$.

20. Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that (f_n) converges uniformly on A to a function $f : A \rightarrow \mathbb{R}$. Then show that f is continuous on A .

21. Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be defined for $n \geq 2$ by :

$$f_n(x) = \begin{cases} n^2 x & , 0 \leq x \leq \frac{1}{n} \\ -n^2(x - 2/n), & \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & , \frac{2}{n} \leq x \leq 1. \end{cases}$$

Show that the limit function is Riemann integrable. Verify whether $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 f(x) dx$.

22. Given $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \pi$, find the value of $\int_0^{\infty} e^{-x^2} dx = \sqrt{\frac{\pi}{2}}$.

23. Show that $\forall p > 0, q > 0 \quad B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$.

(5 × 6 = 30 marks)

Section C

Answer any **two** questions.

Each question carries 10 marks.

24. State and prove Location of roots theorem.

25. State and prove Additivity theorem.

26. Evaluate (a) $\lim_{n \rightarrow \infty} \frac{x^n}{1+x^n}$ for $x \in \mathbb{R}, x \geq 0$. (b) $\lim_{n \rightarrow \infty} \frac{\sin nx}{1+nx}$ for $x \in \mathbb{R}, x \geq 0$.

Discuss about their uniform convergence.

27. (a) Show that $\forall q > -1, \int_0^1 x^q e^{-x} dx$ converges.

(b) Show that $\forall q \leq -1, \int_0^1 x^q e^{-x} dx$ diverges.

(2 × 10 = 20 marks)

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022

Mathematics

MAT 6B 13 [E03]—C PROGRAMMING FOR MATHEMATICAL COMPUTING

(2014 to 2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Section A

*Answer all questions.**Each question carries 1 mark.*

1. C programming language is developed by _____.
2. What is the meaning of the constant `\v` in C ?
3. _____ is an example of a logical operator in C.
4. What is the output of the program ?

```
#include<stdio.h>
#define sqr(i) i*i
main()
{printf("%d", sqr(3), sqr(3+1));}
```

5. What is the difference between a declaration and a definition of a variable in C ?
6. _____ command is used to skip a part of a loop.
7. Write the syntax of *do* statement.
8. Find errors, if any of the statement :

```
if(x + y==z && y > 0)
printf(" ");
```

9. What is the output of the program :

```
#include <stdio.h>
main()
{int a = 3, b = 2; int c = a + b;
printf("%d", &c);}
```

10. What is the general form of declaration of a string variable ?
11. Which of the following is not a storage class specifier in C ?
(a) auto ; (b) register ; (c) static ; (d) extern ; (e) volatile.

12. Which one of the following variable names is NOT valid ?
 (a) go_cart ; (b) go4it ; (c) 4sea ; (d) run6 ; (e) All the above.

(12 × 1 = 12 marks)

Section B

*Answer any nine questions.
 Each question carries 2 marks.*

13. What is the significance of program algorithms ?
14. What are the different steps involved in the execution of C programs ?
15. What are the basic data types associated with C ?
16. Specify the relational operators and their meaning in C.
17. Explain the working of conditional operator in C.
18. What is header file in C ?
19. What do you mean by a garbage value in a variable ?
20. Write a simple C program to find the maximum of three numbers.
21. What is the difference between scanf and getchar ?
22. What is a null statement ? Explain its usefulness.
23. What do you mean by chaining of functions and recursion of functions ?
24. Distinguish between actual and formal arguments.

(9 × 2 = 18 marks)

Section C

*Answer any six questions.
 Each question carries 5 marks.*

25. What is a flow chart ? What are the advantages and drawbacks of flow charts ?
26. What are the rules for identifiers in C ?
27. Explain switch statement in C with syntax and suitable example.
28. Explain the working of for loop in C.
29. Write a C program to reverse a given number.
30. What are the rules apply to a #define statement ?
31. Write a program to print the following output using for loop :

```

      1
    2   2
  3   3   3
4   4   4   4
5   5   5   5   5
  
```

32. Explain the working of *continue* statement using a suitable example.
33. Explain the handling of multidimensional arrays with syntax and suitable example.

(6 × 5 = 30 marks)

Section D

*Answer any two questions.
Each question carries 10 marks.*

34. (a) Write an algorithm and draw the flowchart of the program to check whether a given number is prime or not. (5 marks)
- (b) Explain different storage classes in C. (5 marks)
35. Write an essay on user-defined functions in C.
36. (a) Write a C program to swap two numbers using function. (5 marks)
- (b) Write a C program to reverse a string. (5 marks)

[2 × 10 = 20 marks]